

PATHAHEAD · FREE ROADMAP

Become a Data Engineer from Scratch

Build the pipelines that power every dashboard and ML model. Go from zero to job-ready in about six months, with real pipelines and a data platform capstone.

24

weeks · ~10–12 hrs/wk

5

hands-on projects

4

phases · to production

₹0

free tiers only

A free PathAhead learning resource · adjust the pace to your life — the order matters more than the dates.

THE ONE RULE

Ingest → transform → serve

Data engineering is plumbing at scale — you learn it by moving real data. Every project below builds a pipeline that ingests, transforms, and serves data reliably.

THE ROADMAP

Four phases, step by step

0

Foundations

Weeks 1–5

Python, SQL, and databases — deeply.

WEEK 1

Python for data

Scripting, files, APIs, virtualenvs.

WEEK 2

SQL deep dive

Joins, aggregation, window functions.

WEEK 3

Database design

Relational modeling, normalization, indexes.

WEEK 4

Linux & CLI

Shell, cron, file handling.

WEEK 5

Git & environments

Version control and reproducible setups.

1

Pipelines & ETL

Weeks 6–11

Move and clean data on a schedule.

WEEK 6

ETL/ELT concepts

Extract, transform, load patterns.

WEEK 7

Batch processing

pandas and data transformation at scale.

WEEK 8

Orchestration

Scheduling pipelines with Airflow.

WEEK 9

Ingestion

Pull from APIs, files, and databases.

WEEK 10

Data quality

Validation, tests, and monitoring.

WEEK 11 · PROJECT ►

ETL pipeline

API → clean → load into a database.

2

Big Data & Warehouses

Weeks 12–18

Store and process data at real scale.

WEEK 12**Data warehouses**

Snowflake or BigQuery fundamentals.

WEEK 13**Dimensional modeling**

Star schemas, facts and dimensions.

WEEK 14**Apache Spark**

Distributed processing basics.

WEEK 15**Scaling**

Partitioning, parallelism, big datasets.

WEEK 16**Streaming**

Real-time data with Kafka.

WEEK 17**dbt**

Version-controlled SQL transformations.

WEEK 18 · PROJECT ►**Warehouse + dbt**

Model a warehouse and transform with dbt.

3

Cloud & Scale

Weeks 19–24

Run pipelines in production, in the cloud.

WEEK 19**Cloud data platforms**

AWS or GCP data services.

WEEK 20**Docker for data**

Containerize pipelines.

WEEK 21**Infrastructure basics**

IaC and deployment for data.

WEEK 22**Production orchestration**

Monitoring and alerting on pipelines.

WEEK 23**Performance & cost**

Tuning queries and controlling spend.

WEEK 24 · CAPSTONE ►**Data platform**

An end-to-end ingest-to-serve platform.

BUILD A PORTFOLIO

Five projects that get you hired

Projects beat certificates. Do them in order — each builds on the last. Put every one on GitHub with a clear README: problem → approach → result.

Start here

1 • ETL Script

Pull from a public API, clean it, load it into Postgres.

Orchestration

2 • Airflow Pipeline

A scheduled pipeline that runs and monitors itself.

Warehouse

3 • Warehouse + dbt

Model data in a warehouse and transform it with dbt.

In demand

4 • Streaming Pipeline

Ingest real-time events with Kafka.

Capstone

5 • End-to-End Data Platform

Ingestion → warehouse → transformation → serving layer.

ROLES YOU CAN APPLY FOR

Data Engineer · Analytics Engineer · ETL Developer · Big Data Engineer

TYPICAL STARTING SALARY · INDIA

₹6–14 LPA

LEARN IT

Learning resources

Every item links to the real source (clickable in this PDF).

BOOKS

- [Designing Data-Intensive Applications](#) by Martin Kleppmann
- [Fundamentals of Data Engineering](#) by Joe Reis and Matt Housley
- [The Data Warehouse Toolkit](#) by Ralph Kimball and Margy Ross

COURSES

- [Data Engineering Zoomcamp](#) by DataTalks Club (free)
- [Google Cloud Data Engineering](#) by Coursera
- [Big Data](#) by NPTEL (free)
- [IBM Data Engineering Certificate](#) by Coursera

YOUTUBE

- [Seattle Data Guy](#)
- [Data with Zach](#)
- [freeCodeCamp.org](#)

PRACTICE

- [Kaggle](#)
- [GitHub](#)
- [DataCamp](#)

DO THIS TODAY

Your first step (takes 30 minutes)

1

Install Python + PostgreSQL

Free — your local data engineering workbench.

2

Write one SQL query

Load sample data and query it. The core skill starts here.

3

Build one ETL script

Pull from a public API and save it to a table today.

Job-ready checklist: 3–5 solid projects on GitHub (≥ 1 deployed or documented) · can explain your work in an interview · comfortable with the core tools of the field · a portfolio or profile recruiters can find.

PathAhead — India's premium career guidance platform. This roadmap is a free educational resource; timelines are a guide and everyone learns at a different pace. Referenced third-party resources (Data Engineering Zoomcamp, dbt Learn, DataCamp, freeCodeCamp, Mode SQL Tutorial) are for learning only and are not affiliated with PathAhead.